

State of the Map Europe 2025

Using OpenStreetMap data to create audio only maps

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Soundscape Android project

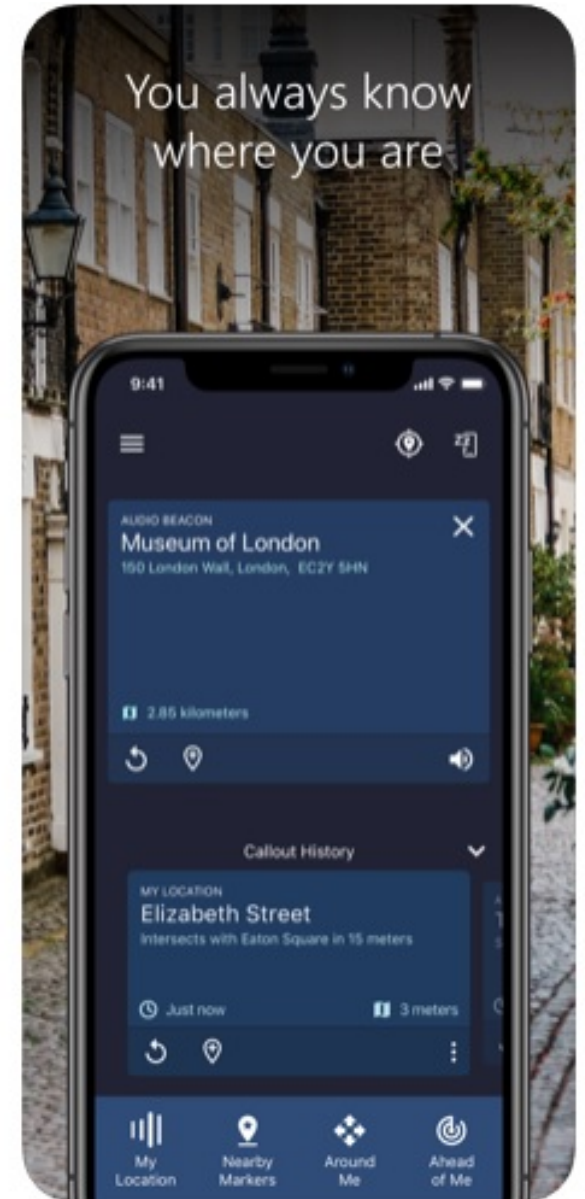
<https://github.com/Scottish-Tech-Army/Soundscape-Android>

All map data used © OpenStreetMap contributors



What is Soundscape?

- An app to help people with visual impairments explore the world by aiding navigation and describing the surroundings.
- Originated as an iPhone app from Microsoft Research that started in 2015 and had 60,000 users.
- Users wear headphones, or glasses with audio
 - Audio beacons
 - Audio description of surroundings
- User interface based on text lists and searches instead of graphical maps



How do audio descriptions compare to graphical maps?

- Graphical maps have a lot of context
- Audio descriptions need to be concise and local
- Describe what the user is currently interested in
- Allow exploration and discovery - enriches knowledge of the area and how you feel about it
- User can add markers to augment OSM data

OpenStreetMap data

- We build our own protomaps planet file and also generate extracts that can be used for offline maps. All using the brilliant [planetiler](#).
- The same MVT tiles are used for the onscreen map and for the audio map.
- Very cost effective and sustainable.
- Customisable maps.

Using map tiles within Soundscape

- MapBox vector tiles are designed for graphical maps.
- Creating tiles involves throwing away a lot of data e.g. simplification of road lines. For the audio descriptions need all of the nodes in roads so that we can detect intersections.
- Lines and polygons need more careful joining at tile edges

What OSM data do we use?

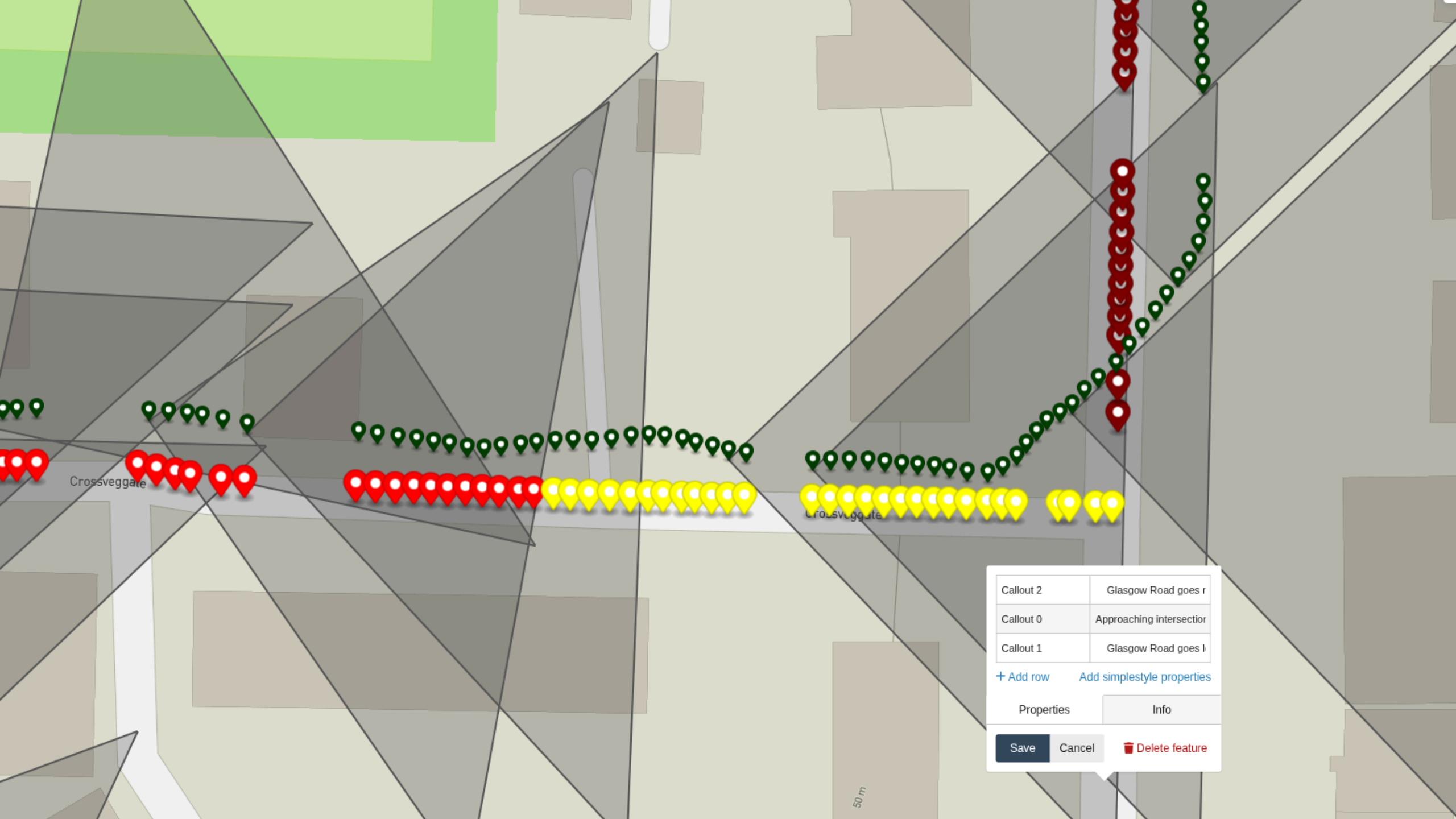
- All path, tracks, roads
- All sorts of Points of Interest
e.g. shops, museums, parks
- Public transit stops
- Accessibility features e.g.
tactile paving
- Building entrances



Map matching to OSM data

Map matching is the process of taking raw GPS locations and translating them into the most likely position of the user. Think of it as "snapping" to the most likely path or roads.

- Map matching for cars is well understood, though it's often done once a journey has been completed.
- It's a harder problem map matching pedestrians and without a crystal ball.
- Our algorithm uses:
 - [An Improved Map-Matching Technique Based on the Fréchet Distance Approach for Pedestrian Navigation Services](#) by Yoonsik Bang, Jiyoung Kim and Kiyun Yu.
 - Dijkstra's algorithm to aid rejection of unlikely paths



Callout 2	Glasgow Road goes r
Callout 0	Approaching intersection
Callout 1	Glasgow Road goes l

[+ Add row](#)[Add simplestyle properties](#)

Properties

Info

Save

Cancel

Delete feature

Describing the surroundings from OSM data

It's only once we've figured out where the user is, and where they've been, that we can accurately describe the surroundings.

- Current location and direction of travel
- Upcoming intersections
- Nearby points of interest

Reproducing graphical map context for unnamed Ways

Lots of paths and ways in the tiles are unnamed and that can lead to poor audio description so we add some of our own context:

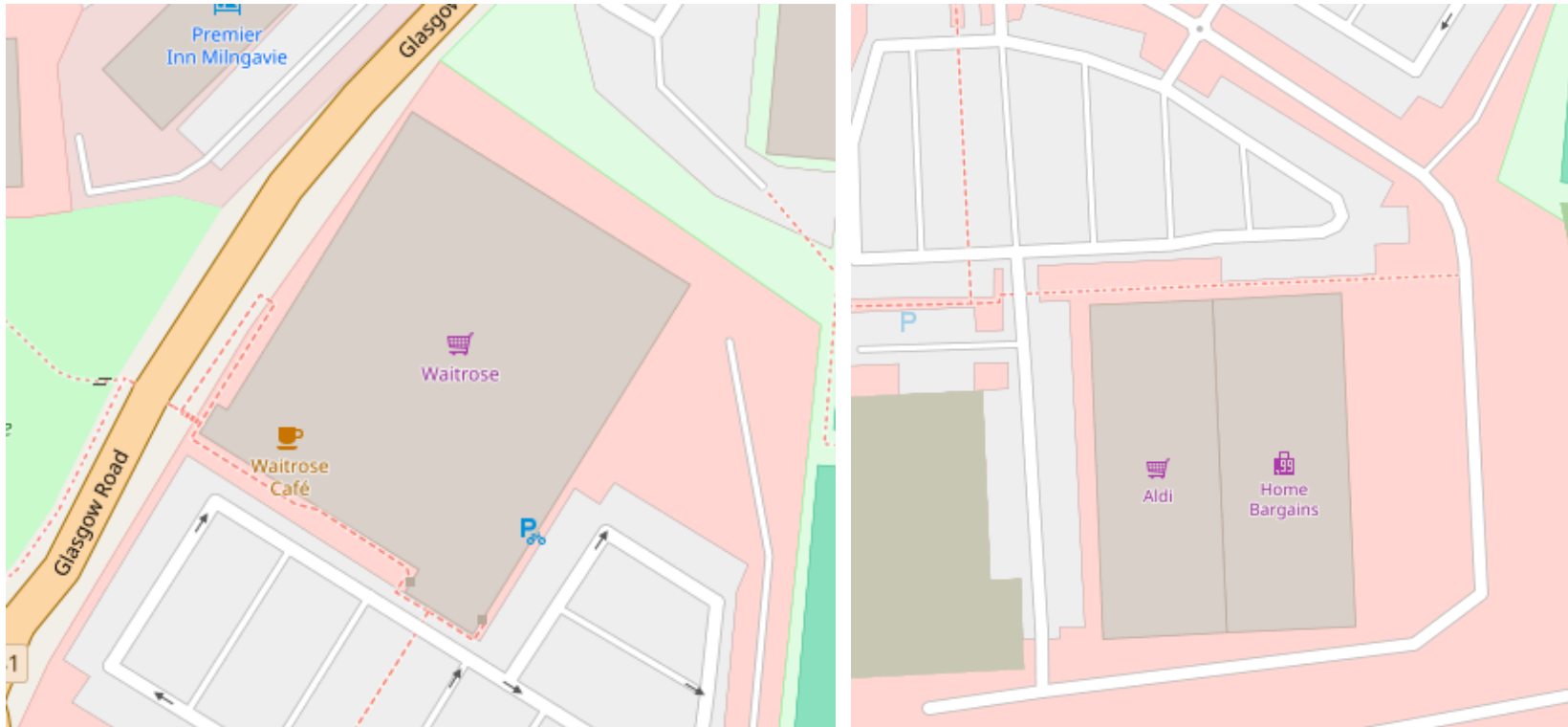
- Adding `destination` key support and making those up e.g. "Path to Moor Road"
- Using nearby points e.g. "Path to James M. Gale Monument"
- Path via steps/bridge/tunnel
- Looking at other adjacent linear features e.g. "Path next to River Dee" or "Path near rail tracks"

Simple vs. Complex road layouts



Entrances **entrance=main**

How do we help people who are blind get the last 15m to their destination?
An important part of this is making sure that their destination is accurate.



Different ways to tag entrances

- Station entrances - `railway=entrance` or `railway=train_station_entrance`
- Subway entrances - `railway=subway_entrance`
- Entrances levels/layers
- Building entrances where there's more than one building polygon

Multistory building and level=*

In high rise urban areas it's useful to include the level for POI.

taipei 101

Way: Breez Nan Shan (513491894)

Version #24

Change wheelchair to yes

Edited 3 months ago by [wheelmap_visitor](#)

Changeset #169195074

Tags

addr:city	台北市
addr:country	TW
addr:district	信義區

Directions and information=guide post

- Several standards for this, most don't contain enough data
- There's a lot of information, and you don't want to force users to listen to it
- Destinations are always within our 2x2 grid so we can't confection them



Summary of actions to aid audio maps

- Consider non-graphical uses when mapping on OSM - is there enough context in the tagging?
- Always map entrances
- Everything that can have a name should be named
- Don't add sidewalks if they can be described using `sidewalk=both`
- Figure out how to deal with fingerposts

Help us

- The app is free and is available on the Play Store in open beta.
- Join or ask for help from the STA.

